Statistical Foundations of ML — Formula Sheet

1 Foundations

Definition 1.1. Sample space S: set of all outcomes. Event: any $A \subseteq S$.

Definition 1.2. A probability measure $\mathbb{P}$ satisfies

- $\mathbb{P}(S) = 1$
- $\mathbb{P}(A) \geq 0$
- $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$ for disjoint A_i.

$$\boxed{\mathbb{P}(A^c) = 1 - \mathbb{P}(A) \qquad \mathbb{P}(\emptyset) = 0 \qquad 0 \leq \mathbb{P}(A) \leq 1}$$

$$\boxed{A \subseteq B \;\Rightarrow\; \mathbb{P}(A) \leq \mathbb{P}(B)}$$

$$\boxed{\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)}$$

Inclusion-Exclusion (3 events)

$$\boxed{\begin{aligned}\mathbb{P}(A \cup B \cup C) = {} & \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ & - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) \\ & + \mathbb{P}(A \cap B \cap C)\end{aligned}}$$

Union bound

$$\boxed{\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) \leq \sum_{i=1}^{n} \mathbb{P}(A_i)}$$

Bonferroni inequalities

With $S_1 = \sum_i \mathbb{P}(A_i)$, $S_2 = \sum_{i<j} \mathbb{P}(A_i \cap A_j)$, $S_3 = \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k)$:

$$\boxed{S_1 - S_2 \;\leq\; \mathbb{P}\Big(\bigcup_i A_i\Big) \;\leq\; S_1 \qquad \mathbb{P}\Big(\bigcup A_i\Big) \leq S_1 - S_2 + S_3}$$

(Stopping inclusion-exclusion after an odd term gives an upper bound; after an even term, a lower bound.)

2 Conditional Probability & Independence

Definition 2.1. $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ for $\mathbb{P}(B) > 0$.

Multiplication (chain) rule

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)}$$

Law of total probability

(B_i partition S)

$$\boxed{\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}$$

Bayes' rule

$$\boxed{\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}}$$

Definition 2.2. A, B independent iff

$$\boxed{\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \quad \Longleftrightarrow \quad \mathbb{P}(A \mid B) = \mathbb{P}(A)}$$

Mutual independence of $A_1, \ldots, A_n$: the product rule holds for *every* subset of the events.

3 Counting & Sampling

$$\boxed{n! = 1 \cdot 2 \cdots n \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}}$$

Choosing k items from n:

	without repetition	with repitition
ordered	$\dfrac{n!}{(n-k)!}$	n^k
unordered	$\binom{n}{k}$	$\binom{n+k-1}{k}$

- Arrangements of n objects: $n!$; circular: $(n-1)!$.
- Binomial theorem: $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.
- Equally likely outcomes: $\mathbb{P}(A) = |A|/|S|$.

4 Random Variables & CDF

Definition 4.1. A random variable X is a real-valued function on the sample space.

Definition 4.2. CDF: $F(x) = \mathbb{P}(X \leq x)$.

$$\boxed{\mathbb{P}(a < X \leq b) = F(b) - F(a) \qquad \mathbb{P}(X > a) = 1 - F(a)}$$

F is non-decreasing, right-continuous, $F(-\infty) = 0$, $F(\infty) = 1$.

5 Discrete Random Variables & PMF

Definition 5.1. PMF: $p(x) = \mathbb{P}(X = x)$.

$$\boxed{p(x) \geq 0, \quad \sum_x p(x) = 1, \quad F(x) = \sum_{t \leq x} p(t)}$$

	$p(x)$	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Bernoulli(p)	$p^x(1-p)^{1-x}$, $x \in \{0,1\}$	p	$p(1-p)$
Binomial(n,p)	$\binom{n}{x}p^x(1-p)^{n-x}$, $x = 0..n$	np	$np(1-p)$
Geometric(p)	$(1-p)^{x-1}p$, $x \geq 1$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$\frac{e^{-\lambda}\lambda^x}{x!}$, $x \geq 0$	λ	λ

- For Geometric RV: $\mathbb{P}(X > k) = (1-p)^k$;
- Memoryless property of Geometric:

$$\boxed{\mathbb{P}(X > m+n \mid X > m) = \mathbb{P}(X > n)}$$

- Binomial: number of successes in n independent Bernoulli(p) trials.

6 Continuous Random Variables & PDF

Definition 6.1. PDF f: $f(x) \geq 0$,

- $\int_{-\infty}^{\infty} f(x)\,dx = 1$, density integrates to 1
- $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx$.

$$\boxed{F(x) = \int_{-\infty}^{x} f(t)\,dt \qquad f(x) = F'(x) \qquad \mathbb{P}(X = x) = 0}$$

	$f(x)$	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
Uniform(a,b)	$\frac{1}{b-a}$, $a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$\lambda e^{-\lambda x}$, $x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ,σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2

- Uniform: $F(x) = \frac{x-a}{b-a}$ for $a \leq x \leq b$.
- Exponential: $F(x) = 1 - e^{-\lambda x}$
- Memoryless property of Exponential:

$$\boxed{\mathbb{P}(X > t) = e^{-\lambda t}}$$

- Normal: Standardize $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$:

$$\boxed{F(x) = \Phi\Big(\frac{x-\mu}{\sigma}\Big) \qquad \Phi(-z) = 1 - \Phi(z)}$$

7 Expectation & Variance

$$\boxed{\mathbb{E}[X] = \sum_x x\,p(x) \qquad \text{or} \qquad \mathbb{E}[X] = \int x f(x)\,dx}$$

$$\boxed{\mathbb{E}[g(X)] = \sum_x g(x)p(x) \qquad \text{or} \qquad \int g(x)f(x)\,dx}$$

$$\boxed{\text{Linearity}: \mathbb{E}\Big[\sum_i X_i\Big] = \sum_i \mathbb{E}[X_i]}$$

$$\boxed{\mathbb{E}[I_A] = \mathbb{P}(A)}$$

Variance

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2}$$

$$\boxed{\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)}$$

If X_i are independent: $\mathrm{Var}(\sum_i X_i) = \sum_i \mathrm{Var}(X_i)$.

8 Random Vectors, Joint CDF & Independence

Definition 8.1. Joint CDF: $F(x,y) = \mathbb{P}(X \le x,\, Y \le y)$.

$$\boxed{F_X(x) = \lim_{y\to\infty} F(x,y) \qquad F_Y(y) = \lim_{x\to\infty} F(x,y)}$$

Definition 8.2. X, Y independent iff

$$\boxed{F(x,y) = F_X(x)\,F_Y(y) \quad \text{for all } x,y}$$

9 Discrete Random Vectors

Definition 9.1. Joint PMF: $p(x,y) = \mathbb{P}(X=x, Y=y)$.

$$\boxed{p(x,y) \ge 0 \qquad \sum_{x,y} p(x,y) = 1}$$

$$\boxed{p_X(x) = \sum_y p(x,y) \qquad p_Y(y) = \sum_x p(x,y)}$$

$$\boxed{p_{X|Y}(x \mid y) = \frac{p(x,y)}{p_Y(y)} \qquad p_Y(y) > 0}$$

$$\boxed{p(x,y) = p_X(x)\,p_Y(y) \quad \text{if independent}}$$

$$\boxed{\mathbb{E}[g(X,Y)] = \sum_{x,y} g(x,y)\,p(x,y)}$$

10 Jointly Continuous Random Vectors

Definition 10.1. Joint PDF $f(x,y) \ge 0$,

- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$
- $\mathbb{P}((X,Y)\in A) = \iint_A f(x,y)\,dx\,dy$

$$\boxed{f(x,y) = \frac{\partial^2 F}{\partial x\,\partial y}}$$

$$\boxed{f_X(x) = \int f(x,y)\,dy \qquad f_Y(y) = \int f(x,y)\,dx}$$

$$\boxed{f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)} \qquad f_Y(y) > 0}$$

$$\boxed{f(x,y) = f_X(x)\,f_Y(y) \quad \text{if independent}}$$

$$\boxed{\mathbb{E}[g(X,Y)] = \iint g(x,y)\,f(x,y)\,dx\,dy}$$

11 Covariance & Correlation

$$\boxed{\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}$$

$$\boxed{\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)}$$

$$\boxed{\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} \qquad -1 \le \rho \le 1}$$

- Independent $\Rightarrow \mathrm{Cov}(X,Y) = 0$